

Redox Reactions

Question1

The oxidation state of chromium in the final product formed in the reaction between KI and acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution is :

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Options:

A.

+2

B.

+3

C.

+4

D.

+6

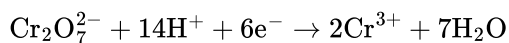
Answer: B

Solution:

In acidified $\text{K}_2\text{Cr}_2\text{O}_7$, chromium is in the +6 oxidation state (in $\text{Cr}_2\text{O}_7^{2-}$).

When KI is added in acidic medium, I^- is oxidised to I_2 and dichromate is reduced to Cr^{3+} .

Reduction half-reaction (NCERT method):



So, chromium in the final product is Cr^{3+} , i.e. oxidation state is +3.

Correct option: B (+3)

Question2

One mole of $\text{Cl}_2(\text{g})$ was passed into 2 L of cold 2 M KOH solution. After the reaction, the concentrations of Cl^- , ClO^- and OH^- are respectively (assume volume remains constant)

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Options:

A.

0.5M, 0.5M, 0.5M

B.

1M, 1M, 1M

C.

0.5M, 0.5M, 1M

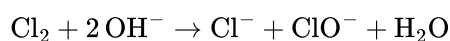
D.

0.75M, 0.75M, 1M

Answer: C

Solution:

In cold KOH, chlorine disproportionates:



Initial moles of OH^- in 2 L of 2 M KOH:

$$n(\text{OH}^-) = 2 \times 2 = 4 \text{ mol}$$

Given 1 mol Cl_2 :

- OH^- consumed = $2 \times 1 = 2 \text{ mol}$
- Cl^- formed = 1 mol
- ClO^- formed = 1 mol
- OH^- remaining = $4 - 2 = 2 \text{ mol}$

Assuming volume stays 2 L:

$$[\text{Cl}^-] = \frac{1}{2} = 0.5 \text{ M}, \quad [\text{ClO}^-] = \frac{1}{2} = 0.5 \text{ M}, \quad [\text{OH}^-] = \frac{2}{2} = 1 \text{ M}$$

Answer: Option C — 0.5 M, 0.5 M, 1 M.

Question3

Given below are two statements:

Statement (I): Oxidising power of halogens decreases in the order $F_2 > Cl_2 > Br_2 > I_2$, which is the basis of "Layer test".

Statement (II): "Layer test" to identify Br_2 and I_2 in aqueous solution involves the oxidation of bromide or iodide into Br_2 or I_2 respectively with Cl_2 , which is a type of displacement redox reaction.

In the light of the above statements, choose the correct answer from the options given below:

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Options:

A.

Both **Statement I** and **Statement II** are true

B.

Both **Statement I** and **Statement II** are false

C.

Statement I is true but **Statement II** is false

D.

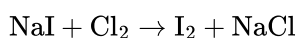
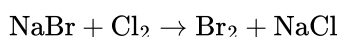
Statement I is false but **Statement II** is true

Answer: A

Solution:

The oxidising power of halogens decreases in the order $F_2 > Cl_2 > Br_2 > I_2$. This means that fluorine is the strongest oxidising agent and iodine is the weakest. Therefore, Cl_2 can easily oxidise bromide and iodide ions to form Br_2 and I_2 respectively. This principle is used in the "Layer test".

In this test, chlorine gas is passed through an aqueous solution containing bromide or iodide ions. A redox (oxidation-reduction) reaction takes place, where chlorine oxidises bromide or iodide ions. The chemical equations are as follows:



So, Cl_2 displaces Br_2 and I_2 from their compounds since chlorine has higher oxidising power. This confirms that both statements (I) and (II) are true, and statement (II) correctly explains statement (I).

Question4

In order to oxidise a mixture of 1 mole each of FeC_2O_4 , $\text{Fe}_2(\text{C}_2\text{O}_4)_3$, FeSO_4 and $\text{Fe}_2(\text{SO}_4)_3$ in acidic medium, the number of moles of KMnO_4 required is

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Options:

A.

3

B.

2

C.

5

D.

7

Answer: B

Solution:

In acidic medium, KMnO_4 acts as a strong oxidising agent.

So, we need to find which species in the given mixture can be oxidised further.

The mixture contains 1 mole each of:

- FeC_2O_4
- $\text{Fe}_2(\text{C}_2\text{O}_4)_3$
- FeSO_4
- $\text{Fe}_2(\text{SO}_4)_3$

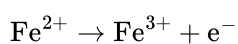
Now let us examine each compound.

1. FeC_2O_4

This is ferrous oxalate, so it contains:

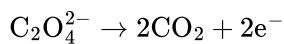
- Fe^{2+} : can be oxidised to Fe^{3+}
- $\text{C}_2\text{O}_4^{2-}$: can be oxidised to CO_2

Oxidation of Fe^{2+}



So, 1 mole of Fe^{2+} gives 1 mole electron.

Oxidation of oxalate



So, 1 mole of $\text{C}_2\text{O}_4^{2-}$ gives 2 moles electrons.

Hence, 1 mole of FeC_2O_4 gives total electrons:

$$1 + 2 = 3$$

2. $\text{Fe}_2(\text{C}_2\text{O}_4)_3$

This contains:

- Fe^{3+} : already in higher oxidation state, so no oxidation
- 3 oxalate ions: each can be oxidised

Each oxalate ion gives 2 electrons, so 3 oxalates give:

$$3 \times 2 = 6$$

Thus, 1 mole of $\text{Fe}_2(\text{C}_2\text{O}_4)_3$ gives 6 electrons.

3. FeSO_4

This contains:

- Fe^{2+} : can be oxidised to Fe^{3+}
- SO_4^{2-} : sulphur is already at maximum oxidation state +6, so no oxidation

Thus, 1 mole of FeSO_4 gives:

1 electron

4. $\text{Fe}_2(\text{SO}_4)_3$

This contains:

- Fe^{3+} : cannot be oxidised further here
- SO_4^{2-} : cannot be oxidised further

So, electrons given:

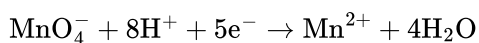
0

Total electrons released

$$3 + 6 + 1 + 0 = 10$$

Electrons accepted by KMnO_4 in acidic medium

In acidic medium:



So, 1 mole of KMnO_4 accepts 5 moles of electrons.

Therefore, moles of KMnO_4 required:

$$\frac{10}{5} = 2$$

Final answer

2

So, the correct option is **Option B**.

Question 5

200 cc of $x \times 10^{-3}$ M potassium dichromate is required to oxidise 750 cc of 0.6 M Mohr's salt solution in acidic medium.

Here $x =$ _____ .

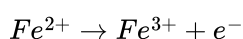
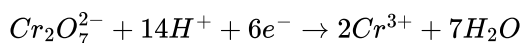
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Answer: 375

Solution:

In acidic medium, dichromate oxidises Fe^{2+} to Fe^{3+} .

Half-reactions (NCERT method):



So, 1 mole of $Cr_2O_7^{2-}$ oxidises 6 moles of Fe^{2+} .

Step 1: Moles of Fe^{2+} in Mohr's salt solution

Volume = 750 cc = 0.75 L, Molarity = 0.6 M

$$n(Fe^{2+}) = 0.75 \times 0.6 = 0.45 \text{ mol}$$

Step 2: Moles of $K_2Cr_2O_7$ required

$$n(K_2Cr_2O_7) = \frac{0.45}{6} = 0.075 \text{ mol}$$

Step 3: Use given dichromate solution data

Volume = 200 cc = 0.2 L, Molarity = $x \times 10^{-3}$ M

$$n = 0.2 \times x \times 10^{-3} = 2 \times 10^{-4}x$$

Equate moles:

$$2 \times 10^{-4}x = 0.075$$

$$x = \frac{0.075}{2 \times 10^{-4}} = 375$$

$$x = 375$$

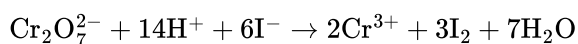
Question6

X and Y are the number of electrons involved, respectively during the oxidation of I^- to I_2 and S^{2-} to S by acidified $K_2Cr_2O_7$. The value of $X + Y$ is _____ .

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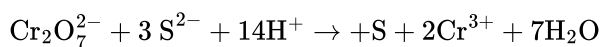
Answer: 12

Solution:



In this reaction, $6I^-$ ions get oxidised to $3I_2$.

Each I^- loses 1 electron during oxidation, so total electrons lost (in moles) = $X = 6$.



Here, $3S^{2-}$ ions get oxidised to $3S$.

Each S^{2-} loses 2 electrons to become S, so total electrons lost (in moles) = $Y = 3 \times 2 = 6$.

Therefore, $x + y = 6 + 6 = 12$.

Question7

500 mL of 1.2 M KI solution is mixed with 500 mL of 0.2MKMnO₄ solution in basic medium. The liberated iodine was titrated with standard 0.1MNa₂ S₂O₃ solution in the presence of starch indicator till the blue color disappeared. The volume (in L) of Na₂ S₂O₃ consumed is _____. (Nearest integer)

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Answer: 3

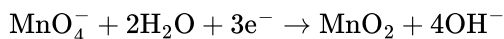
Solution:

In basic medium, KMnO_4 is reduced to MnO_2 and I^- is oxidised to I_2 .

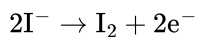
Step 1: Balance the redox reaction in basic medium

Half-reactions:

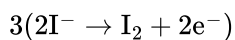
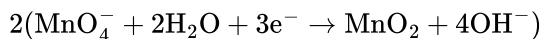
Reduction:



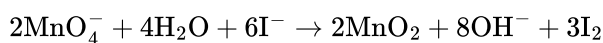
Oxidation:



LCM of electrons = 6:



Adding:



So,

- 2 mol MnO_4^- produce 3 mol I_2 .

Step 2: Calculate moles of reactants

Moles of KI:

$$n(\text{I}^-) = 0.5 \times 1.2 = 0.6 \text{ mol}$$

Moles of KMnO_4 :

$$n(\text{MnO}_4^-) = 0.5 \times 0.2 = 0.1 \text{ mol}$$

From the balanced equation:

$$1 \text{ mol MnO}_4^- \text{ requires } 3 \text{ mol I}^-$$

So 0.1 mol MnO_4^- requires 0.3 mol I^- , which is available (0.6 mol).

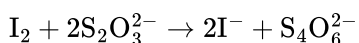
Hence, KMnO_4 is the limiting reagent.

Step 3: Moles of I_2 liberated

$$n(\text{I}_2) = \frac{3}{2} \times 0.1 = 0.15 \text{ mol}$$

Step 4: Titration with thiosulfate

Reaction:



So,

$$n(\text{S}_2\text{O}_3^{2-}) = 2 \times 0.15 = 0.30 \text{ mol}$$

Given $\text{Na}_2\text{S}_2\text{O}_3$ is 0.1 M:

$$V = \frac{n}{M} = \frac{0.30}{0.1} = 3.0 \text{ L}$$

Answer (nearest integer)

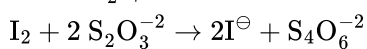
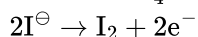
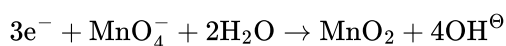
Question8

500 mL of 0.2M MnO_4^- solution in basic medium when mixed with 500 mL of 1.5 M KI solution, oxidises iodide ions to liberate molecular iodine. This liberated iodine is then titrated with a standard $x\text{M}$ thiosulphate solution in presence of starch till the end point. If 300 mL of thiosulphate was consumed, then the value of x is ____.

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Answer: 1

Solution:



gm equivalent of MnO_4^-

$$= 0.2 \times \frac{500}{1000} \times 3 = 0.3 \text{ (Limiting reagent)}$$

gm equivalent of KI

$$= 1.5 \times \frac{500}{1000} \times 1 = 0.75$$

gm equivalent of MnO_4^-

= gm equivalent of hypo

$$0.3 = x \times \frac{300}{1000} \times 1$$

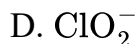
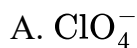
$$x = 1$$

Question9

The species which does not undergo disproportionation reaction is :

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Options:



Answer: A

Solution:

Oxidation States:

In ClO_4^- (perchlorate), chlorine is in the +7 oxidation state.

In ClO_3^- (chlorate), chlorine is in the +5 oxidation state.

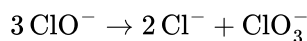
In ClO_2^- (chlorite), chlorine is in the +3 oxidation state.

In ClO^- (hypochlorite), chlorine is in the +1 oxidation state.

Disproportionation Reaction:

A disproportionation reaction is one in which a species simultaneously undergoes oxidation and reduction. For this to occur, the element must be in an intermediate oxidation state such that it can be oxidized to a higher state and reduced to a lower one.

In ClO^- and ClO_2^- , chlorine is in lower oxidation states (+1 and +3, respectively), making them susceptible to disproportionation. For example, hypochlorite can disproportionate in basic solution as follows:



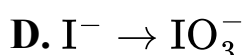
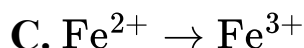
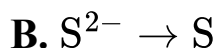
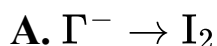
In ClO_3^- , chlorine is in an intermediate oxidation state (+5) that, under certain conditions, can undergo disproportionation.

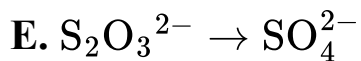
However, in ClO_4^- , chlorine is in its highest possible oxidation state (+7) and cannot be oxidized further. Since disproportionation requires one part of the species to be oxidized and the other reduced, ClO_4^- is thermodynamically stable and does not undergo disproportionation.

ClO_4^- (perchlorate ion) does not undergo disproportionation.

Question10

Which of the following oxidation reactions are carried out by both $\text{K}_2\text{Cr}_2\text{O}_7$ and KMnO_4 in acidic medium?





Choose the correct answer from the options given below :

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Options:

A. A, B and C Only

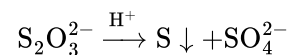
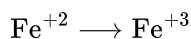
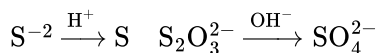
B. C, D and E Only

C. B, C and D Only

D. A, D and E Only

Answer: A

Solution:



Question11

Match the List - I with List - II

	List - I (Redox Reaction)		List - II (Type of Redox Reaction)
(A)	$\text{CH}_{4(g)} + 2\text{O}_{2(g)} \xrightarrow{\Delta} \text{CO}_{2(g)} + 2\text{H}_2\text{O}_{(l)}$	(I)	Disproportionation reaction
(B)	$2\text{NaH}_{(s)} \xrightarrow{\Delta} 2\text{Na}_{(s)} + \text{H}_{2(g)}$	(II)	Combination reaction
(C)	$\text{V}_2\text{O}_{5(s)} + 5\text{Ca}_{(s)} \xrightarrow{\Delta} 2\text{V}_{(s)} + 5\text{CaO}_{(s)}$	(III)	Decomposition reaction
(D)	$2\text{H}_2\text{O}_{2(aq)} \xrightarrow{\Delta} 2\text{H}_2\text{O}_{(l)} + \text{O}_{2(g)}$	(IV)	Displacement reaction

Choose the correct answer from the options given below:

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Options:

A. A-III, B-IV, C-I, D-II

B. A-IV, B-I, C-II, D-III

C. A-II, B-III, C-I, D-IV

D. A-II, B-III, C-IV, D-I

Answer: D

Solution:

(A) Combustion of hydrocarbon

(B) Decomposition into gaseous product.

(C) Displacement of 'V' by 'Ca' atom.

(D) Disproportionation of $\text{H}_2\text{O}_2^{-1}$ into O^{-2} and O^0 oxidation states.

Question12

Given below are two statements:

Statement I: In the oxalic acid vs KMnO_4 (in the presence of dil H_2SO_4) titration the solution needs to be heated initially to 60°C , but no heating is required in Ferrous ammonium sulphate (FAS) vs KMnO_4 titration (in the presence of dil H_2SO_4)

Statement II: In oxalic acid vs KMnO_4 titration, the initial formation of MnSO_4 takes place at high temperature, which then acts as catalyst for further reaction. In the case of FAS vs KMnO_4 , heating oxidizes Fe^{2+} into Fe^{3+} by oxygen of air and error may be introduced in the experiment.

In the light of the above statements, choose the correct answer from the options given below

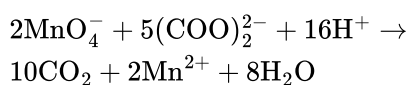
Options:

- A. Statement I is false but Statement II is true
- B. Statement I is true but Statement II is false
- C. Both Statement I and Statement II are true
- D. Both Statement I and Statement II are false

Answer: C

Solution:

For the titration: Oxalic acid v/s KMnO_4

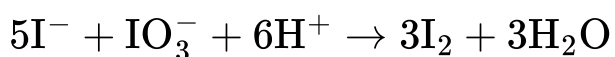


This reaction is slow at room temperature, but becomes fast at 60°C . Manganese(II) ions catalyse the reaction; thus, the reaction is autocatalytic; once manganese(II) ions are formed, it becomes faster and faster.

The titration of FAS v/s KMnO_4 do not require heating because at higher temperature the oxidation of Fe^{+2} to Fe^{+3} by atmospheric O_2 will be prominent.

Question13

**0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solutions.
According to equation**



Identify the correct statements :

- (A) 200 mL of KI solution reacts with 0.004 mol of KIO_3
- (B) 200 mL of KI solution reacts with 0.006 mol of H_2SO_4
- (C) 0.5 L of KI solution produced 0.005 mol of I_2
- (D) Equivalent weight of KIO_3 is equal to $\left(\frac{\text{Molecular weight}}{5} \right)$

Choose the correct answer from the options given below :

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Options:

A.

(C) and (D) only

B.

(A) and (B) only

C.

(B) and (C) only

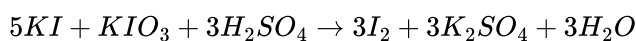
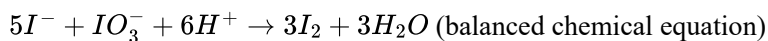
D.

(A) and (D) only

Answer: D

Solution:

Molarity of KI = 0.1 M



(A) Volume given (KI) = 200 mL = 0.2 L

Molarity (KI) = 0.1 M or mol L⁻¹

The molarity formula is used here to calculate the number of moles of K.I.

$$\text{Molarity, } M = \frac{\text{number of moles}}{\text{volume in L}}$$

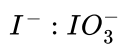
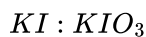
So, number of moles = Molarity × Volume in L

For the given volume of KI,

$$\text{number of moles} = 0.1 \text{ mol L}^{-1} \times 0.2 \text{ L}$$

$$= 0.02 \text{ mol}$$

The stoichiometric ratio between KI and KIO₃ is



$$5 : 1$$

$$1 : \frac{1}{5}$$

For one mol I⁻, $\frac{1}{5}$ mol IO₃⁻ is used.

So, for 0.2 mol I⁻,

$$\text{Moles of } IO_3^- = 0.02 \times \frac{1}{5} = 0.004 \text{ mol}$$

The statement (A) is correct.

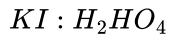
(B) Volume given (KI) = 200 mL = 0.2 L

Molarity (KI) = 0.1 M

number of moles of KI (I^-) = Molarity $\times$ Volume (L)

$$= 0.1 \text{ mol L}^{-1} \times 0.2 \text{ L} = 0.02 \text{ mol}$$

The stoichiometric ratio between KI and H_2HO_4 is



$$5 : 3$$

$$1 : \frac{3}{5} \text{ \{3 moles } H_2SO_4 \text{ can give } 6H^+\}$$

$$I^-, \frac{3}{5}$$

For one mol I^- , $\frac{3}{5}$ mol H_2SO_4 is used.

So, for 0.02 mol I^- ,

$$\text{moles of } H_2SO_4 = 0.02 \times \frac{3}{5} = 0.012 \text{ mol}$$

This statement is not correct.

(C) Volume given (KI) = 0.5 L

Molarity (KI) = 0.1 M

Number of moles of KI = Molarity $\times$ Volume

$$= 0.1 \text{ mol L}^{-1} \times 0.5 \text{ L} = 0.05 \text{ mol}$$

The stoichiometric ratio between I^- and I_2 is



$$5 : 3$$

$$1 : \frac{3}{5}$$

For 1 mol I^- , $\frac{3}{5}$ mol I_2 is produced.

$$\text{So, for 0.05 mol, moles of } I_2 = 0.05 \times \frac{3}{5} = 0.03 \text{ mol}$$

This statement is not correct.

$$(D) \text{ Equivalent weight of } KIO_3 = \frac{\text{Molecular weight}}{5}$$

This statement is correct.

$$\text{Equivalent weight} = \frac{\text{Molecular weight}}{\text{Valency factor} \rightarrow \text{number of electrons}}$$

Here, valency factor = 5

Each KIO_3 molecule gains 5 electrons in this reaction.

KIO_3 acts as the oxidizing agent, gains electrons from the iodide ion in KI.

When KIO_3 is reduced to I_2 each molecule gains 5 electrons. So, the valency factor is 5 and equivalent weight is lower than molecular weight.

This statement is correct.

Correct statements are A and D.

Question14

Some CO_2 gas was kept in a sealed container at a pressure of 1 atm and at 273 K. This entire amount of CO_2 gas was later passed through an aqueous solution of $\text{Ca}(\text{OH})_2$. The excess unreacted $\text{Ca}(\text{OH})_2$ was later neutralized with 0.1 M of 40 mL HCl. If the volume of the sealed container of CO_2 was x , then x is _____ cm^3 (nearest integer). [Given : The entire amount of CO_2 (g) reacted with exactly half the initial amount of $\text{Ca}(\text{OH})_2$ present in the aqueous solution.]

JEE Main 2025 (Online) 22nd January Morning Shift

Answer: 45

Solution:

Let moles of $\text{CO}_2 = n$

moles of $\text{Ca}(\text{OH})_2$

total initially = $2n$

excess $\text{Ca}(\text{OH})_2 = n$

gm equivalent of $\text{Ca}(\text{OH})_2 = \text{gm equivalent of HCl}$

$$n \times 2 = 0.1 \times \frac{40}{1000} \times 1$$

$$n = 2 \times 10^{-3}$$

$$\text{Volume of } \text{CO}_2 = 2 \times 10^{-3} \times 22400 = 44.8 \text{ cm}^3$$

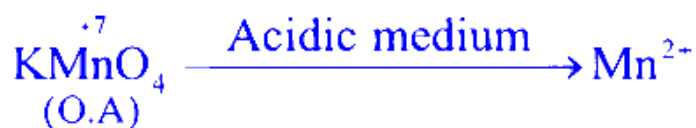
Question15

KMnO_4 acts as an oxidising agent in acidic medium. 'X' is the difference between the oxidation states of Mn in reactant and product. 'Y' is the number of 'd' electrons present in the brown red precipitate formed at the end of the acetate ion test with neutral ferric chloride. The value of $X + Y$ is _____.

JEE Main 2025 (Online) 4th April Morning Shift

Answer: 10

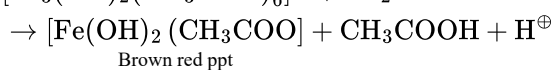
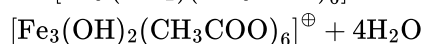
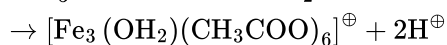
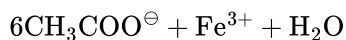
Solution:



X is difference in oxidation state.

$$7 - 2 = 5$$

$$\text{So } X = 5$$



$$\text{So } Y = 5$$

$$X + Y = 5 + 5 = 10$$

Question16

5g of NaOH was dissolved in deionized water to prepare a 450 mL stock solution. What volume (in mL) of this solution would be required to prepare 500 mL of 0.1M solution?

**Given : Molar Mass of Na, O and H is 23, 16 and 1 gmol⁻¹ respectively
[24-Jan-2023 Shift 1]**

Answer: 180

Solution:

$$M = \frac{5}{40} \times \frac{1000}{450}$$

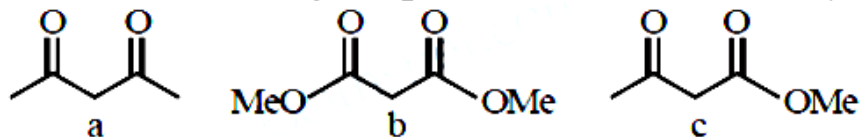
$$M_1 V_1 = M_2 V_2$$

$$\left(\frac{5}{40} \times \frac{1000}{450} \right) \times V_1 = 0.1 \times 500$$

$$V_1 = 180$$

Question17

Which will undergo deprotonation most readily in basic medium?



[24-Jan-2023 Shift 2]

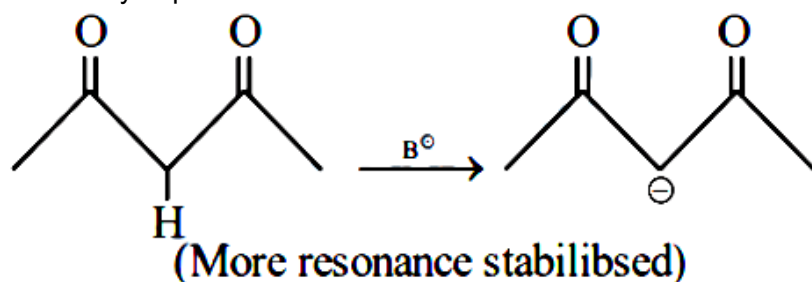
Options:

- A. a only
- B. c only
- C. Both a and c
- D. b only

Answer: A

Solution:

Most easily deprotonation



Question18

The density of a monobasic strong acid (Molar mass 24.2g mol) is 1.21 kg L. The volume of its solution required for the complete neutralization of 25 mL of 0.24M NaOH is _____ x 10⁻² mL (Nearest integer)

[25-Jan-2023 Shift 1]

Answer: 12

Solution:

$$\begin{aligned}
 &\text{millimole of NaOH} = 0.24 \times 25 \\
 &\therefore \text{millimole of acid} = 0.24 \times 25 \\
 &\Rightarrow \text{mass of acid} = 0.24 \times 25 \times 24.2 \text{ mg} \\
 &\text{for pure acid,} \\
 &V = \frac{W}{d}; (d = 1.21 \text{ kg/L} = 1.21 \text{ g/ml}) \\
 &\therefore V = \frac{0.24 \times 25 \times 24.2}{1.12} \times 10^{-3} \\
 &= 120 \times 10^{-3} \text{ ml} \\
 &= 12 \times 10^{-2} \text{ ml}
 \end{aligned}$$

Question19

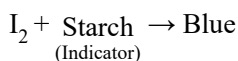
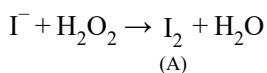
An indicator ' X ' is used for studying the effect of variation in concentration of iodide on the rate of reaction of iodide ion with H_2O_2 at room temp. The indicator ' X ' forms blue colored complex with compound ' A ' present in the solution. The indicator ' X ' and compound ' A ' respectively are
[29-Jan-2023 Shift 2]

Options:

- A. Starch and iodine
- B. Methyl orange and H_2O_2
- C. Starch and H_2O_2
- D. Methyl orange and iodine

Answer: A

Solution:



Question20

The volume of HCl , containing 73 g L^{-1} , required to completely neutralise NaOH obtained by reacting 0.69 g of metallic sodium with water, is _____ mL. (Nearest Integer)

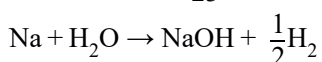
(Given : molar Masses of Na, Cl, O, H are 23, 35.5, 16 and 1 g mol^{-1})

respectively)
[29-Jan-2023 Shift 2]

Answer: 15

Solution:

$$\text{Mole of Na} = \frac{0.69}{23} = 3 \times 10^{-2}$$



By using POAC

$$\text{Moles of NaOH} = 3 \times 10^{-2}$$

NaOH reacts with HCl

No. of equivalent of NaOH = No. of equivalent of HCl

$$3 \times 10^{-2} \times 1 = \frac{73}{36.5} \times V(\text{ in L}) \times 1$$

$$V = 1.5 \times 10^{-2} \text{L}$$

Volume of HCl = 15 ml.

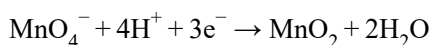
Question21

The number of electrons involved in the reduction of permanganate to manganese dioxide in acidic medium is _____.

[30-Jan-2023 Shift 1]

Answer: 3

Solution:

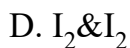
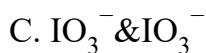
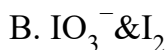
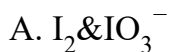


Question22

KMnO₄ oxidises I⁻ in acidic and neutral/faintly alkaline solution, respectively to

[30-Jan-2023 Shift 2]

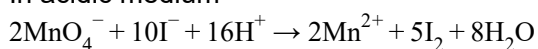
Options:



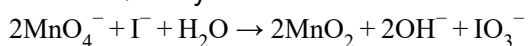
Answer: A

Solution:

In acidic medium



In neutral/faintly alkaline solution



Question23

25 mL of an aqueous solution of KCl was found to require 20 mL of 1M AgNO_3 solution when titrated using K_2CrO_4 as an indicator. What is the depression in freezing point of KCl solution of the given concentration?

_____ (Nearest integer).

(Given : $K_f = 2.0\text{K kg mol}^{-1}$)

Assume

1) 100% ionization and

2) density of the aqueous solution as 1gmL^{-1}

[1-Feb-2023 Shift 1]

Answer: 3

Solution:

At equivalence point, mmole of KCl = mmole of AgNO_3 = 20 mmole

Volume of solution = 25 ml

Mass of solution = 25 gm

Mass of solvent

= 25 - mass of solute

$$= 25 - [20 \times 10^{-3} \times 74.5]$$

$$= 23.51 \text{ gm}$$

$$\text{Molality of KCl} = \frac{\text{mole of KCl}}{\text{mass of solvent in kg}}$$

$$= \frac{20 \times 10^{-3}}{23.51 \times 10^{-3}} = 0.85$$

$$i \text{ of KCl} = 2(100\% \text{ ionisation})$$

$$\Delta T_f = i \times K_f \times m$$

$$= 2 \times 2 \times 0.85$$

$$= 3.4$$

$$\approx 3$$

Question24

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : An aqueous solution of KOH when for volumetric analysis, its concentration should be checked before the use.

Reason (R) : On aging, KOH solution absorbs atmospheric CO_2 .

In the light of the above statements, choose the correct answer from the options given below.

[1-Feb-2023 Shift 2]

Options:

- A. (A) is not correct but (R) is correct
- B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- C. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- D. (A) is correct but (R) is not correct

Answer: C

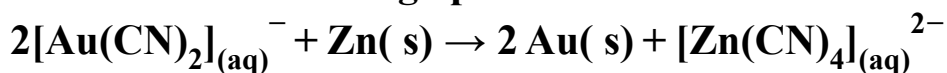
Solution:

KOH absorb CO_2

So its concentration should be checked.

Question25

Which of the following options are correct for the reaction



A. Redox reaction

B. Displacement reaction

C. Decomposition reaction

D. Combination reaction

Choose the correct answer from the options given below:

[6-Apr-2023 shift 1]

Options:

A. A and B only

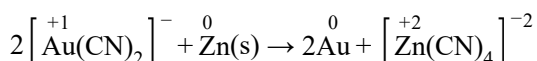
B. A only

C. C and D only

D. A and D only

Answer: A

Solution:



Zn displaced Au^+

Reduction and Oxidation both are taking place.

Question26

Strong reducing and oxidizing agents among the following, respectively, are

[6-Apr-2023 shift 1]

Options:

A. Ce^{4+} and Eu^{2+}

B. Ce^{4+} and Tb^{4+}

C. Ce^{3+} and Ce^{4+}

D. Eu^{2+} and Ce^{4+}

Answer: D

Question27

During the reaction of permanganate with thiosulphate, the change in oxidation of manganese occurs by value of 3 . Identify which of the below medium will favour the reaction.

[6-Apr-2023 shift 2]

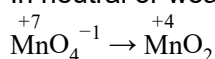
Options:

- A. aqueous acidic
- B. aqueous neutral
- C. both aqueous acidic and neutral
- D. both aqueous acidic and faintly alkaline.

Answer: B

Solution:

In neutral or weakly alkaline solution oxidation state of Mn changes by 3 unit



Question28

The volume of 0.02M aqueous HBr required to neutralize 10.0 mL of 0.01M aqueous Ba(OH)₂ is (Assume complete neutralization)

[6-Apr-2023 shift 2]

Options:

- A. 2.5 mL
- B. 5.0 mL
- C. 10.0 mL
- D. 7.5 mL

Answer: C

Solution:

$$\begin{aligned} N_1 v_1 &= N_2 v_2 \\ \Rightarrow 0.02 v_1 &= 0.02 \times 10 \\ \Rightarrow v_1 &= 10 \text{ ml} \end{aligned}$$

Question29



What is the value of x ?

[8-Apr-2023 shift 1]

Options:

A. 12

B. 10

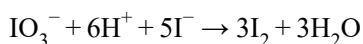
C. 2

D. 6

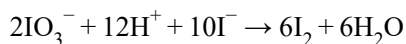
Answer: B

Solution:

n factor of IO_3^- and I^- in the given redox reaction are 5 and 1 respectively. Therefore, IO_3^- will always react in the molar ratio 1 : 5 to get I_2 .



To get 6 molar I_2 , multiple equation by 2



So, $x = 10$

Question30

Given below are two statement :

Statement I : In redox titration, the indicators used are sensitive to change in pH of the solution.

Statement II : In acid-base titration, the indicators used are sensitive to change in oxidation potential.

In the light of the above statement, choose the most appropriate answer from the options given below

[8-Apr-2023 shift 2]

Options:

A. Both statement I and Statement II are incorrect

B. Statement I is incorrect but Statement II is correct

C. Statement I is correct but Statement II is incorrect

D. Both Statement I and Statement II are correct

Answer: A

Solution:

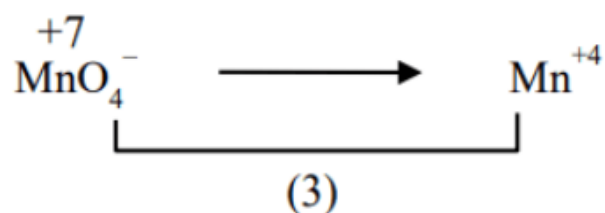
Fact

Question31

In alkaline medium, the reduction of permanganate anion involves a gain of _____ electrons.
[10-Apr-2023 shift 2]

Answer: 3

Solution:

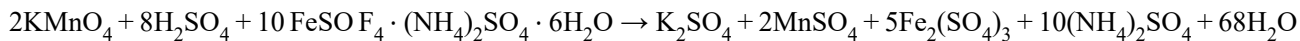


Question32

KMnO₄ is titrated with ferrous ammonium sulphate hexahydrate in presence of dilute H₂SO₄. Number of water molecules produced for 2 molecules of KMnO₄ is _____.
[13-Apr-2023 shift 1]

Answer: 68

Solution:



On the basis of above equation, 68 molecules of water will be produced from 2 molecules of KMnO_4 .

Question33

20 mL of calcium hydroxide was consumed when it was reacted with 10 mL of unknown solution of H_2SO_4 . Also 20 mL standard solution of 0.5 M HCl containing 2 drops of phenolphthalein was titrated with calcium hydroxide, the mixture showed pink colour when burette displayed the value of 35.5 mL whereas the burette showed 25.5 mL initially. The concentration of H_2SO_4 is _____ M. (Nearest integer)
[13-Apr-2023 shift 1]

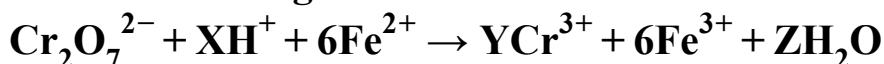
Answer: 1

Solution:

$$\begin{aligned} \text{miliequivalent of } \text{Ca}(\text{OH})_2 &= \text{miliequivalent of } \text{H}_2\text{SO}_4 \\ M_1 \times 2 \times 20 &= M_2 \times 2 \times 10 \\ 2M_1 &= M_2 \\ \text{miliequivalent of } \text{HCl} &= \text{miliequivalent of } \text{Ca}(\text{OH})_2 \\ 20 \times 0.5 &= 10 \times M_1 \times 2 \\ M_1 &= 0.5\text{M} \\ \text{Concentration of } \text{H}_2\text{SO}_4 &= M_2 = 2M_1 \\ &= 2 \times 0.5 \\ &= 1\text{M} \end{aligned}$$

Question34

See the following chemical reaction:

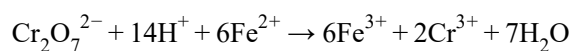


The sum of X, Y and Z is _____

[13-Apr-2023 shift 2]

Answer: 23

Solution:



$$x = 14$$

$$y = 2$$

$$z = 7$$

$$\text{Hence } (x + y + z) = 14 + 2 + 7 = 23$$

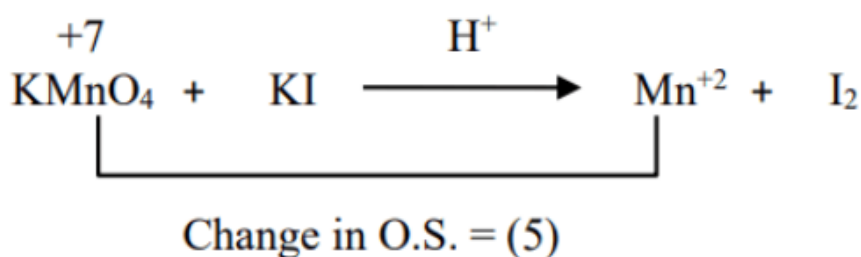
Question35

The total change in the oxidation state of manganese involved in the reaction of KMnO_4 and potassium iodide in the acidic medium is _____.

[15-Apr-2023 shift 1]

Answer: 5

Solution:



Question36

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Physical properties of isotopes of hydrogen are different.

Reason : Mass difference between isotopes of hydrogen is very large.

In the light of the above statements, choose the correct answer from the options given below:

[10-Apr-2023 shift 2]

Options:

A. Both A and R are true but R is NOT the correct explanation of A

B. A is false but R is true

C. A is true but R is false

D. Both A and R are true and R is the correct explanation of A

Answer: D

Solution:

The Physical properties of isotope of Hydrogen are different due to Large mass difference

Question37

A 0.166g sample of an organic compound was digested with conc. H_2SO_4 and then distilled with NaOH. The ammonia gas evolved was passed through 50.0 mL of 0.5N H_2SO_4 . The used acid required 30.0 mL of 0.25N NaOH for complete neutralization. The mass percentage of nitrogen in the organic compound is
[24-Jun-2022-Shift-1]

Answer: 63

Solution:

$$\text{Millimoles of used acid} = \frac{30 \times 0.25}{2}$$

$$\text{Millimoles of } \text{NH}_3 = 30 \times 0.25 = 7.5$$

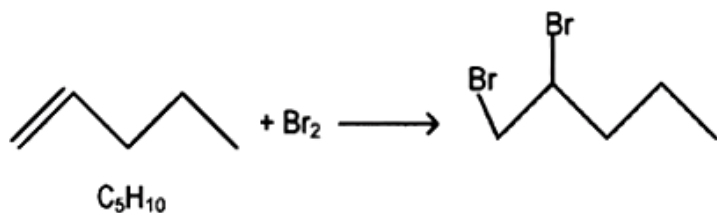
$$\text{Mass \% of nitrogen} = \frac{7.5}{0.166} \times 10^{-3} \times 14 \times 100 \approx 63\%$$

Question38

Number of grams of bromine that will completely react with 5.0g of pent-1-ene is $\times 10^{-2}$ g. (Atomic mass of Br = 80g/mol) [Nearest Integer]
[25-Jun-2022-Shift-1]

Answer: 1143

Solution:



Molar mass of $C_5H_{10} = 12 \times 5 + 10 = 70$ gm

Given mass of $C_5H_{10} = 5$ gm

$$\therefore \text{Moles of } C_5H_{10} = \frac{5}{70}$$

From reaction,

1 mole of C_5H_{10} reacts with 1 mole of Br_2

$$\therefore \frac{5}{70} \text{ moles of } C_5H_{10} \text{ reacts with } \frac{5}{70} \text{ moles of } Br_2$$

$$\therefore \text{Reacted } Br_2 = \frac{5}{70} \times 160 \text{ gm}$$

$$= 11.428 \text{ gm}$$

$$= 1142.8 \times 10^{-2} \text{ gm}$$

$$\simeq 1143 \times 10^{-2} \text{ gm}$$

Question39

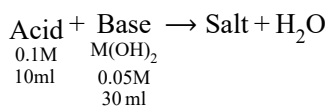
The neutralization occurs when 10 mL of 0.1M acid 'A' is allowed to react with 30 mL of 0.05M base $M(OH)_2$. The basicity of the acid 'A' is

[M is a metal]

[25-Jun-2022-Shift-2]

Answer: 3

Solution:



at equivalence point equivalent of acid = equivalent of base

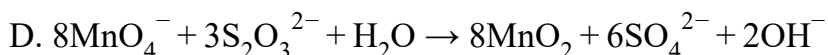
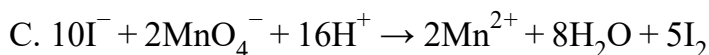
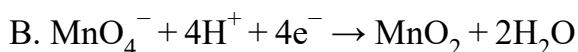
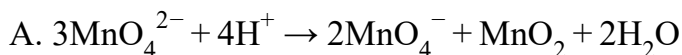
$$0.1 \times 10 \times n = 30 \times 0.05 \times 2$$

$$n = 3$$

Question40

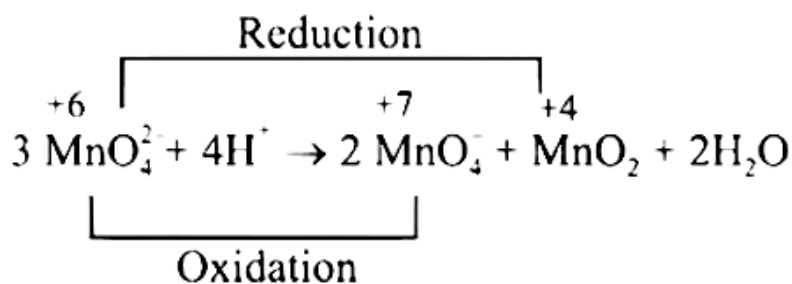
**Which one of the following is an example of disproportionation reaction?
[26-Jun-2022-Shift-2]**

Options:



Answer: A

Solution:



MnO_4^{2-} is an intermediate oxidation state and is converted into compounds having higher and lower oxidation states.

Question41

A 2.0g sample containing MnO_2 is treated with HCl liberating Cl_2 . The Cl_2 gas is passed into a solution of KI and 60.0 mL of 0.1 $\text{MNa}_2\text{S}_2\text{O}_3$ is required to titrate the liberated iodine. The percentage of MnO_2 in the sample is ____

(Nearest integer)

[Atomic masses (in u)

$\text{Mn} = 55$; $\text{Cl} = 35.5$; $\text{O} = 16$, $\text{I} = 127$, $\text{Na} = 23$, $\text{K} = 39$, $\text{S} = 32$]

[28-Jun-2022-Shift-1]

Answer: 13

Solution:

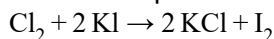
First Step :



Here 1 mol of MnO_2 produce 1 mol of Cl_2

$$\therefore \text{Mole ratio of } n_{\text{MnO}_2} : n_{\text{Cl}_2} = 1 : 1$$

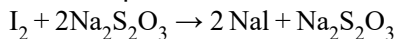
Second Step :



Here, 1 mol of Cl_2 produce 1 mol of I_2

$$\text{Mole ratio of } n_{\text{Cl}_2} : n_{\text{I}_2} = 1 : 1$$

Third Step :



1 mol of I_2 react with 2 mol of $\text{Na}_2\text{S}_2\text{O}_3$

$$\text{Mole ratio of } n_{\text{I}_2} : n_{\text{Na}_2\text{S}_2\text{O}_3} = 1 : 2$$

Given $\text{Na}_2\text{S}_2\text{O}_3$ is 60 mL of 0.1M

$$\therefore \text{Number of moles of } \text{Na}_2\text{S}_2\text{O}_3$$

$$= V(\text{ in L }) \times M (\text{Molarity})$$

$$= \frac{60}{1000} \times 0.1$$

$$= 0.006 \text{ mol}$$

$$\therefore \text{Number of moles of } \text{I}_2$$

$$= \frac{1}{2}(0.006)$$

$$= 0.003$$

$$\therefore \text{Moles of } \text{MnO}_2 = 0.003 \text{ (as mole ratio of } \text{MnO}_2 \text{ and } \text{Cl}_2 = 1 : 1 \text{)}$$

$$\text{Molar mass of } \text{MnO}_2 = 55 + 32 = 87$$

$$\therefore \text{Mass of } \text{MnO}_2 = 0.003 \times 87 = 0.261 \text{ gm}$$

$$\text{Given } \text{MnO}_2 = 2\text{g}$$

$$\therefore \% \text{ of } \text{MnO}_2 = \frac{0.261}{2} \times 100 = 13\%$$

Question42

0.01 KMnO_4 solution was added to 20.0 mL of 0.05M Mohr s salt solution through a burette. The initial reading of 50 mL burette is zero. The volume of KMnO_4 solution left in the burette after the end point is _____mL. (nearest integer)

[28-Jun-2022-Shift-2]

Answer: 30

Solution:

Meq of oxidizing agent = Meq of reducing agent

$$(M \times V \times n_F)_{\text{KMnO}_4} = (M \times V \times n_F)_{\text{Mohr' salt}}$$

$$0.01 \times 20 \times 5 = 0.05 \times V \times 1$$

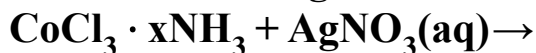
Volume required = 20 ml

Since initial volume of KMnO_4 in burette is 50 ml.

Hence volume of KMnO_4 left in the burette after end point is 30 ml.

Question43

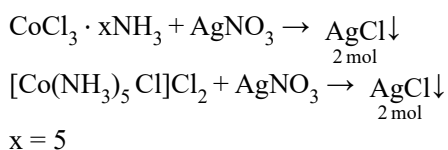
For the reaction given below:



If two equivalents of AgCl precipitate out, then the value of x will be _____
[29-Jun-2022-Shift-2]

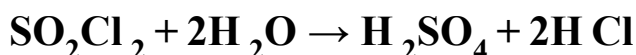
Answer: 5

Solution:



Question44

SO_2Cl_2 on reaction with excess of water results into acidic mixture



16 moles of NaOH is required for the complete neutralisation of the resultant acidic mixture. The number of moles of SO_2Cl_2 used is:

[25-Jul-2022-Shift-1]

Options:

A. 16

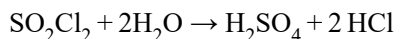
B. 8

C. 4

D. 2

Answer: C

Solution:



Moles of NaOH required for complete neutralization of resultant acidic mixture = 16 moles

And 1 mole of SO_2Cl_2 produced 4 moles of H^+ .

$\therefore$ Moles of SO_2Cl_2 used will be $= \frac{16}{4} = 4$ moles

Question45

In base vs. acid titration, at the end point methyl orange is present as [25-Jul-2022-Shift-2]

Options:

- A. quinonoid form
- B. heterocyclic form
- C. phenolic form
- D. benzenoid form

Answer: A

Question46

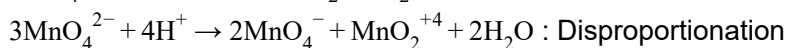
Which of the given reactions is not an example of disproportionation reaction? [26-Jul-2022-Shift-1]

Options:

- A. $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
- B. $2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_3 + \text{HNO}_2$
- C. $\text{MnO}_4^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{MnO}_2 + 2\text{H}_2\text{O}$
- D. $3\text{MnO}_4^{2-} + 4\text{H}^+ \rightarrow 2\text{MnO}_4^- + \text{MnO}_2 + 2\text{H}_2\text{O}$

Answer: C

Solution:



Question47

The dark purple colour of KMnO_4 disappears in the titration with oxalic acid in acidic medium. The overall change in the oxidation number of manganese in the reaction is :

[26-Jul-2022-Shift-1]

Options:

A. 5

B. 1

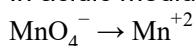
C. 7

D. 2

Answer: A

Solution:

In acidic medium,



change in ox. no. = 5

Question48

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Phenolphthalein is a pH dependent indicator, remains colourless in acidic solution and gives pink colour in basic medium.

Reason R: Phenolphthalein is a weak acid. It doesn't dissociate in basic medium.

In the light of the above statements, choose the most appropriate answer from the options given below.

[26-Jul-2022-Shift-2]

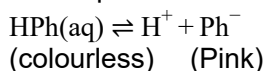
Options:

- A. Both A and R are true and R is the correct explanation of A.
- B. Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.
- C. A is true but R is false.
- D. A is false but R is true.

Answer: C**Solution:**

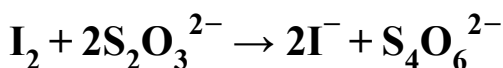
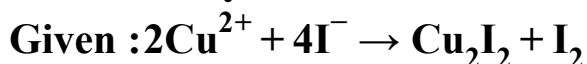
Phenolphthalein is a pH dependent indicator. It is a weak acid which is colourless in the acidic solution but gives pink colour in basic medium. The pink colour is due to its conjugate form. Therefore, assertion (A) is true but Reason (R) is false.

Phenolphthalein dissociate in basic medium

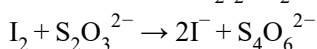
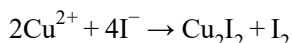
**Question49**

20 mL of 0.02M hypo solution is used for the titration of 10 mL of copper sulphate solution, in the presence of excess of KI using starch as an indicator.

The molarity of Cu^{2+} is found to be _____ $\times 10^{-2}\text{M}$ [nearest integer]



[26-Jul-2022-Shift-2]

Answer: 4**Solution:**

Milliequivalents of hypo solution = $0.02 \times 20 = 0.4$

Milliequivalents of Cu^{2+} in 10 mL solution =

Milliequivalents of I_2 = Milliequivalents of hypo = 0.4

Millimoles of Cu^{2+} ions in 10 mL = 0.4

Molarity of Cu^{2+} ions = $\frac{0.4}{10} = 0.04\text{M}$

= $4 \times 10^{-2}\text{M}$

Question50

20 mL of 0.02M $\text{K}_2\text{Cr}_2\text{O}_7$ solution is used for the titration of 10 mL of Fe^{2+} solution in the acidic medium.

The molarity of Fe^{2+} solution is _____ $\times 10^{-2}\text{M}$. (Nearest Integer)
[27-Jul-2022-Shift-1]

Answer: 24

Solution:

Eq. of $\text{K}_2\text{Cr}_2\text{O}_7$ = Eq. of Fe^{2+}

$\Rightarrow (\text{Molarity} \times \text{volume} \times \text{n.f.}) \text{ of } \text{K}_2\text{Cr}_2\text{O}_7 = (\text{molarity} \times \text{volume} \times \text{n.f.}) \text{ of } \text{Fe}^{2+} \Rightarrow 0.02 \times 20 \times 6 = M \times 10 \times 1 \Rightarrow M = 0.24$

$\Rightarrow \text{Molarity} = 24 \times 10^{-2}$

Question51

Given below are two statements :

One is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Permanganate titrations are not performed in presence of hydrochloric acid.

Reason R : Chlorine is formed as a consequence of oxidation of hydrochloric acid.

In the light of the above statements, choose the correct answer from the options given below

[28-Jul-2022-Shift-2]

Options:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is NOT the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A

Solution:



HCl is not used in the process of titration because it reacts with the (KMnO_4) that is used in the process and gets oxidized.

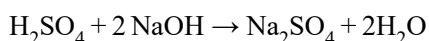
Question52

2L of 0.2M H_2SO_4 is reacted with 2L of 0.1M NaOH solution, the molarity of the resulting product Na_2SO_4 in the solution is _____ millimolar. (Nearest integer)

[28-Jul-2022-Shift-2]

Answer: 25

Solution:



0.4 mol	0.2 mol	—
0.3 mol	—	0.1 mol

Molarity of Na_2SO_4 is $\frac{0.1}{4} = 0.025\text{M}$
= 25 mM

Question53

A compound 'X' is a weak acid and it exhibits colour change at pH close to the equivalence point during neutralization of NaOH with CH_3COOH .

Compound 'X' exists in ionized form in basic medium. The compound 'X' is

[29-Jul-2022-Shift-2]

Options:

A. methyl orange

B. methyl red

C. phenolphthalein

D. erichrome Black T

Answer: C

Solution:

Phenolphthalein is weak acid give colour in basic medium.

Question54

Dichromate ion is treated with base, the oxidation number of Cr in the product formed is

[26 Feb 2021 Shift 1]

Answer: 6

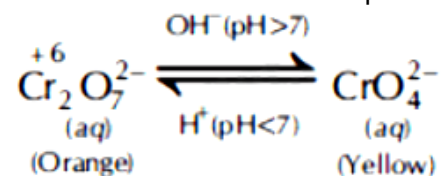
Solution:

In basic medium dichromate ion ($\text{Cr}_2\text{O}_7^{2-}$)

changes in chromate ion (CrO_4^{2-}). Oxidation state of Cr in CrO_4^{2-} is +6.

$\Rightarrow \text{CrO}_4^{2-} = x + 4(-2) = -2$ or $x = +6$

Dichromate and chromate equilibrium depends on pH of the medium as

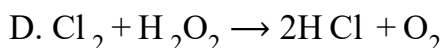
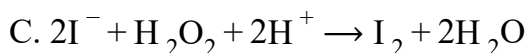
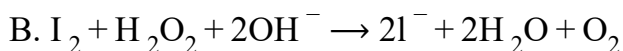
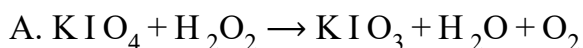


Question55

Which of the following equation depicts the oxidising nature of H_2O_2 ?

[25 Feb 2021 Shift 1]

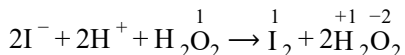
Options:



Answer: C

Solution:

Oxidation involves increase in oxidation number and reduction involves decrease in oxidation number.



In this reaction, H_2O_2 oxidises I^{-} to I_2 and itself gets reduced to H_2O , so the reaction depicts oxidising nature of H_2O_2 . While in other reactions H_2O_2 does not oxidise KIO_4 , I_2 and Cl_2 .

Question56

**In basic medium CrO_4^{2-} oxidises $\text{S}_2\text{O}_3^{2-}$ to form SO_4^{2-} and itself changes into $\text{Cr}(\text{OH})_4^{-}$. The volume of 0.154M CrO_4^{2-} required to react with 40mL of 0.25M $\text{S}_2\text{O}_3^{2-}$ is mL
(Rounded off to the nearest integer).
[25 Feb 2021 Shift 1]**

Answer: 173

Solution:

Given, Molarity of $\text{CrO}_4^{2-}(\text{M}_1) = 0.154\text{M}$

Molarity of $\text{S}_2\text{O}_3^{2-}(\text{M}_2) = 0.25\text{M}$

Volume of $\text{S}_2\text{O}_3^{2-}(\text{V}_2) = 40\text{mL}$

Volume of $\text{CrO}_4^{2-}(\text{V}_1) = ?$ $\text{CrO}_4^{2-} + \text{S}_2\text{O}_3^{2-} \rightarrow \text{SO}_4^{2-} + \text{Cr}(\text{OH})_4^{-}$

Gram equivalent of $\text{CrO}_4^{2-} = \text{Gram equivalent of } \text{S}_2\text{O}_3^{2-}$

$$N_1 V_1 = N_2 V_2$$

Normality = Molarity $\times$ n factor

n for Cr = 6 - 3 = 3

n for S $\Rightarrow \text{S}_2\text{O}_3^{2-} \rightarrow 2\text{SO}_4^{2-} + 8\text{e}^{-}$

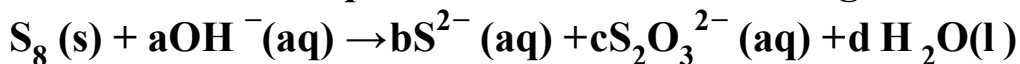
= 8

$$0.154 \times 3 \times V_1 = 0.25 \times 40 \times 8$$

$$V_1 = 173\text{mL}$$

Question57

The reaction of sulphur in alkaline medium is given below:

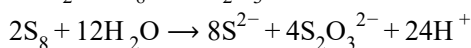
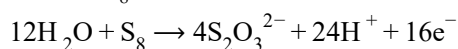
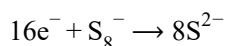


The values of 'a' is _____. (Integer answer)

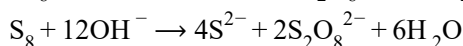
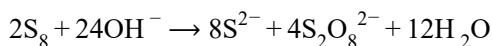
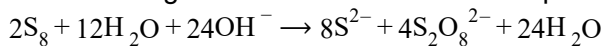
[24feb2021shift1]

Answer: 12

Solution:



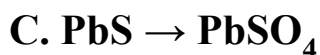
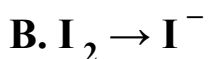
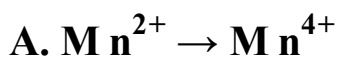
For balancing in basic medium add OH^- equal to H^+ .



$\therefore a = 12$.

Question58

In basic medium, H_2O_2 exhibits which of the following reactions ?



Choose the most appropriate answer from the options given below.

[18 Mar 2021 Shift 2]

Options:

A. A and C

B. Only A

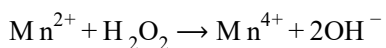
C. B Only

D. A and B

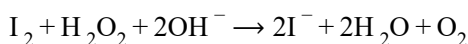
Answer: D

Solution:

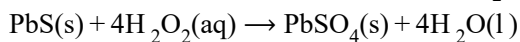
In basic medium, oxidising action of H_2O_2



In basic medium, reducing action of H_2O_2



In acidic medium, oxidising action of H_2O_2



Thus, in basic medium H_2O_2 exhibits reaction 'A' and 'B'

Hence, correct option is (d).

Question 59

Given below are two statements.

Statement I Potassium permanganate on heating at 573K forms potassium manganate.

Statement II Both potassium permanganate and potassium manganate are tetrahedral and paramagnetic in nature. In the light of the above statements, choose the most appropriate answer from the options given below

[17 Mar 2021 Shift 1]

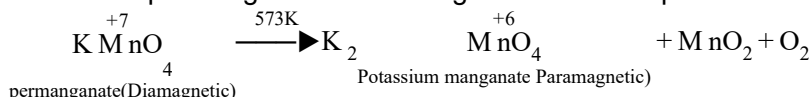
Options:

- A. Statement I is true but statement II is false
- B. Both statement I and statement II are true
- C. Statement I is false but statement II is true
- D. Both statement I and statement II are false

Answer: A

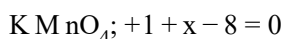
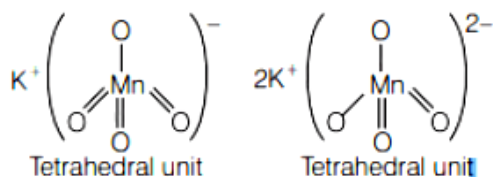
Solution:

Potassium permanganate on heating at 573K forms potassium manganate.

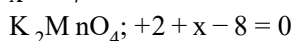


Both potassium permanganate and potassium manganate are tetrahedral.

Manganese has four sigma bonds forming tetrahedral structure.

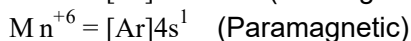
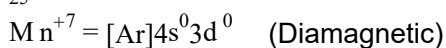
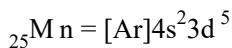


$$x = +7$$



$$x = +6$$

So, Mn is present in +7 oxidation state in KMnO_4 and +6 oxidation state in K_2MnO_4 .



So, potassium permanganate is diamagnetic and potassium manganate is paramagnetic.

Statement I is true but statement II is false.

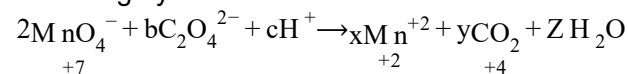
Question60

$2\text{MnO}_4^- + \text{bC}_2\text{O}_4^{2-} + \text{cH}^+ \rightarrow \text{xMn}^{2+} + \text{yCO}_2 + \text{zH}_2\text{O}$ If the above equation is balanced with integer coefficients, the value of c is
[16 Mar 2021 Shift 1]

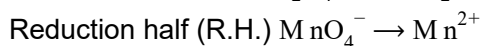
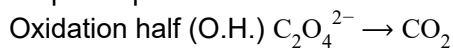
Answer: 16

Solution:

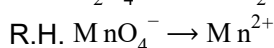
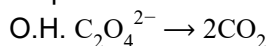
Balancing by half-reaction method



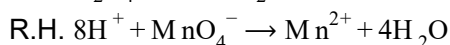
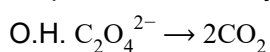
Step 1 Separate two half-reactions



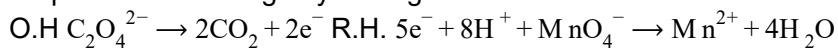
Step 2 Balance all the atoms except oxygen and 'H'.



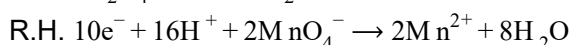
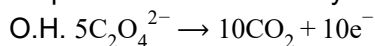
Step 3 Balance 'O' by adding H_2O and 'H' by adding H^+ .



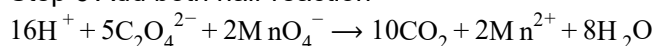
Step 4 Balance charge by adding electron



Step 5 Balance electron by multiplying O.H. by '5' and R.H. by '2'.



Step 6 Add both half-reaction

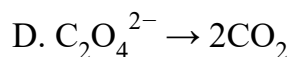
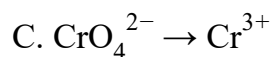
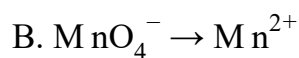
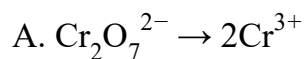


So, c = 16

Question61

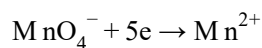
**Identify the process in which change in the oxidation state is five :
[25 Jul 2021 Shift 2]**

Options:



Answer: B

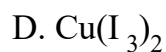
Solution:



Question62

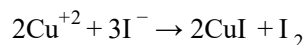
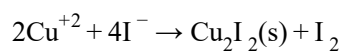
**Cu^{2+} salt reacts with potassium iodide to give
[20 Jul 2021 Shift 2]**

Options:



Answer: A

Solution:



Question63

**The species given below that does NOT show disproportionation reaction is :
[20 Jul 2021 Shift 1]**

Options:



Answer: A

Solution:

In BrO_4^- , Br is in highest oxidation state (+7), So it cannot oxidise further hence it cannot show disproportionation reaction.

Question64

**The sum of oxidation states of two silver ions in $[\text{Ag}(\text{NH}_3)_2][\text{Ag}(\text{CN})_2]$ complex is
[1 Sep 2021 Shift 2]**

Answer: 2

Solution:

$[\text{Ag}(\text{NH}_3)_2][\text{Ag}(\text{CN})_2]$ complex dissociates into $[\text{Ag}(\text{NH}_3)_2]^+$ and $[\text{Ag}(\text{CN})_2]^-$.

Oxidation of Ag in $[\text{Ag}(\text{NH}_3)_2]^+$

$$\text{Ag} + 0 \times 2 = +1$$

$$\text{Ag} = +1$$

Oxidation state of Ag in $[\text{Ag}(\text{CN})_2]^-$

$$\text{Ag} + (-1) \times 2 = -1$$

$$\text{Ag} - 2 = -1$$

$$\Rightarrow \text{Ag} = +1$$

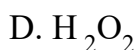
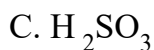
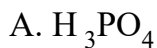
$\therefore$ Sum of oxidation states of two silver ions in

$[\text{Ag}(\text{NH}_3)_2][\text{Ag}(\text{CN})_2]$ complex is 2.

Question65

The compound that cannot act both as oxidising and reducing agent is:
[Jan. 09, 2020(I)]

Options:



Answer: A

Solution:

In H_3PO_4 oxidation state of P is +5, which cannot be oxidised further to a higher oxidation state. Hence, it cannot act as reducing agent.

Question66

The redox reaction among the following is:
[Jan. 07, 2020 (II)]

Options:

A. formation of ozone from atmospheric oxygen in the presence of sunlight

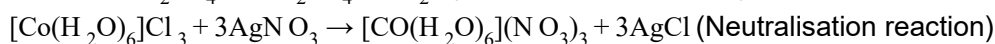
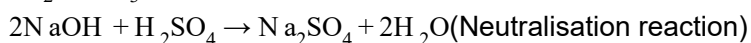
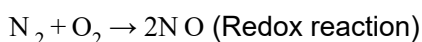
B. reaction of $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$ with AgNO_3

C. reaction of H_2SO_4 with NaOH

D. combination of dinitrogen with dioxygen at 2000K

Answer: D

Solution:



Question67

Oxidation number of potassium in K_2O , K_2O_2 and KO_2 , respectively, is:
[Jan. 07, 2020 (I)]

Options:

A. $+2$, $+1$ and $+\frac{1}{2}$

B. $+1$, $+1$ and $+1$

C. $+1$, $+4$ and $+2$

D. $+1$, $+2$ and $+4$

Answer: B

Solution:

$$K_2O : 2x - 2 = 0 \Rightarrow x = +1$$

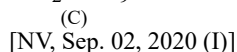
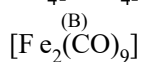
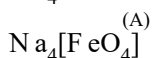
$$K_2O_2 : 2x - 2 = 0 \Rightarrow x = +1$$

$$KO_2 : x - 1 = 0 \Rightarrow x = +1$$

Thus, potassium shows $+1$ state in all its oxides, superoxides and peroxides.

Question68

The oxidation states of iron atoms in compounds (A), (B) and (C), respectively, are x, y and z. The sum of x, y and z is _____.



Answer: 6

Solution:

The oxidation states of iron in these compounds will be -

$$\text{In A, } x + 5(-1) + (-1) = -4 \Rightarrow x = +2$$

$$\text{In B, } y + 4(-2) = -4 \Rightarrow y = +4$$

In C, $z = 0$

The sum of oxidation states will be $= 4 + 2 + 0 = 6$.

Question 69

The oxidation states of transition metal atoms in $K_2Cr_2O_7$, $KMnO_4$ and K_2FeO_4 , respectively, are x , y and z . The sum of x , y and z is _____,
[NV, Sep. 02, 2020 (II)]

Answer: 19

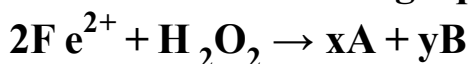
Solution:

Compound	Oxidation state of transition element
(i) $K_2Cr_2O_7$	$x = +6$
(ii) $KMnO_4$	$y = +7$
(iii) K_2FeO_4	$z = +6$

So, $(x + y + z) = 6 + 7 + 6 = 19$

Question 70

Consider the following equations :



(in basic medium)



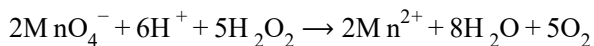
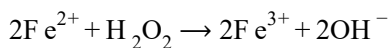
(in acidic medium)

The sum of the stoichiometric coefficients x , y , x' , y' and z' for products A, B, C, D and E, respectively, is _____.

[NV, Sep. 04, 2020 (II)]

Answer: 19

Solution:

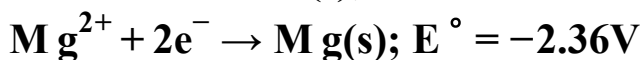
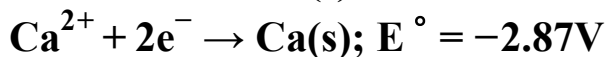
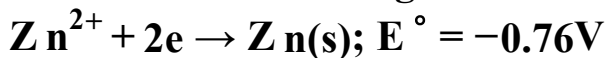


$$\therefore x = 2, y = 2, x' = 2, y' = 8, z' = 5$$

$$\therefore x + y + x' + y' + z' = 19$$

Question 71

Consider the following reduction processes:



The reducing power of the metals increases in the order:

[Jan.10,2019(I)]

Options:

A. $\text{Ca} < \text{Zn} < \text{Mg} < \text{Ni}$

B. $\text{Ni} < \text{Zn} < \text{Mg} < \text{Ca}$

C. $\text{Zn} < \text{Mg} < \text{Ni} < \text{Ca}$

D. $\text{Ca} < \text{Mg} < \text{Zn} < \text{Ni}$

Answer: B

Solution:

Higher the oxidation potential, higher will be the reducing power. So, the order of reducing behaviour is:
 $\text{Ca} > \text{Mg} > \text{Zn} > \text{Ni}$

Question 72

In the reaction of oxalate with permanganate in acidic medium, the number of electrons involved in producing one molecule of CO_2 is:

[Jan. 10,2019 (II)]

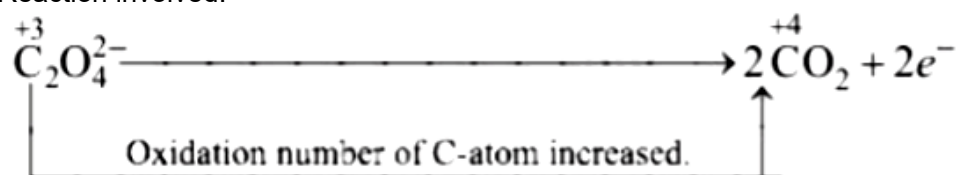
Options:

- A. 1
- B. 10
- C. 2
- D. 5

Answer: A

Solution:

Reaction involved:



∴ The number of electrons involved in producing one mole of CO₂ is 1 .

Question73

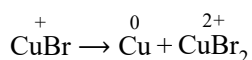
**An example of a disproportionation reaction is:
[April 12, 2019(I)]**

Options:

- A. $2\text{MnO}_4^- + 10\text{I}^- + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 5\text{I}_2 + 8\text{H}_2\text{O}$
- B. $2\text{NaBr} + \text{Cl}_2 \rightarrow 2\text{NaCl} + \text{Br}_2$
- C. $2\text{KMnO}_4 \rightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$
- D. $2\text{CuBr} \rightarrow \text{CuBr}_2 + \text{Cu}$

Answer: D

Solution:



It is an example of disproportionation reaction, as Cu undergoes both oxidation and reduction.

Question74

In order to oxidise a mixture of one mole of each of FeC_2O_4 , $\text{Fe}_2(\text{C}_2\text{O}_4)_3$, FeSO_4 and $\text{Fe}_2(\text{SO}_4)_3$ in acidic medium, the number of moles of KMnO_4 required is :

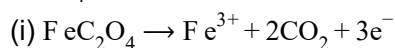
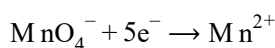
[April 8, 2019 (I)]

Options:

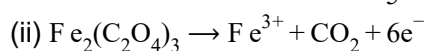
- A. 2
- B. 1
- C. 3
- D. 1.5

Answer: A

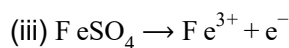
Solution:



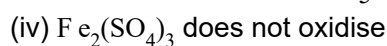
1 mole of FeC_2O_4 reacts with $\frac{3}{5}$ mole of acidified KMnO_4



1 mole of $\text{Fe}_2(\text{C}_2\text{O}_4)_3$ reacts with $\frac{6}{5}$ moles of KMnO_4



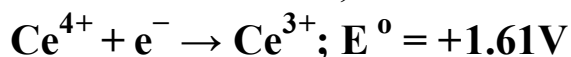
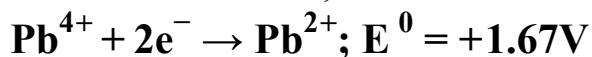
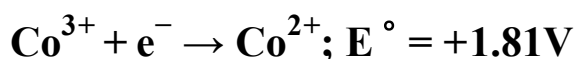
1 mole of FeSO_4 react with $\frac{1}{5}$ moles of KMnO_4



$$\therefore \text{Total moles required} = \frac{3}{5} + \frac{6}{5} + \frac{1}{5} = 2$$

Question 75

Given:

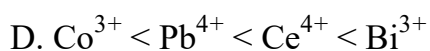
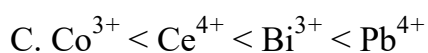
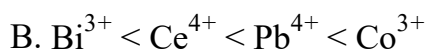


oxidizing power of the species will increase in the order:

[April 12, 2019(I)]

Options:

- A. $\text{Ce}^{4+} < \text{Pb}^{4+} < \text{Bi}^{3+} < \text{Co}^{3+}$



Answer: B

Solution:

Higher the reduction potential, higher will be oxidising power. So,
 $\text{Bi}^{3+} < \text{Ce}^{4+} < \text{Pb}^{4+} < \text{Co}^{3+}$

Question76

Given that $E^\circ_{\text{O}_2/\text{H}_2\text{O}} = +1.23\text{V}$;

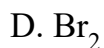
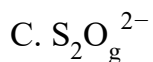
$$E_{\text{S}_2\text{O}_8^{2-}/\text{SO}_4^{2-}} = 2.05\text{V}$$

$$E_{\text{Br}_2/\text{Br}^\circ} = +1.09\text{V}$$

$$E_{\text{Au}^{3+}/\text{Au}}^\circ = +1.4\text{V}$$

The strongest oxidising agent is:
[April 8, 2019 (I)]

Options:



Answer: C

Solution:

More positive is the reduction potential stronger is the oxidising agent. Reduction potential is maximum for $\text{S}_2\text{O}_8^{2-}$, therefore, it is the strongest oxidising agent amongst the given species.

Question77

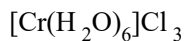
The oxidation states of Cr in $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$, $[\text{Cr}(\text{C}_6\text{H}_6)_2]$, and $\text{K}_2[\text{Cr}(\text{CN})_2(\text{O})_2(\text{O}_2)(\text{NH}_3)]$ respectively are :
[2018]

Options:

- A. +3, +4, and +6
- B. +3, +2, and +4
- C. +3, 0, and +6
- D. +3, 0, and +4

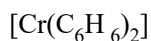
Answer: C

Solution:

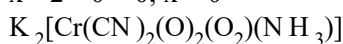


$$\Rightarrow x + 6 \times 0 + (-1) \times 3 = 0$$

$$\Rightarrow x = +3$$



$$x + 2 \times 0 = 0; x = 0$$



[Here $(\text{O})_2$ is OX O, $(2x - 2)$ and (O_2) is per OX O, $(1x - 2)$]

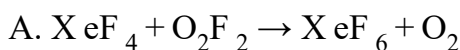
$$2 \times 1 + x + 2 \times (-1) + 2 \times (-2) + (-2) + 0 = 0$$

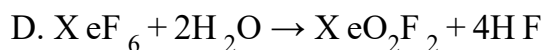
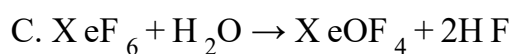
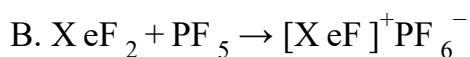
$$x = +6$$

Question78

Which of the following reactions is an example of a redox reaction?
[2017]

Options:

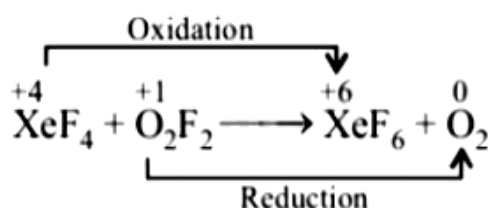




Answer: A

Solution:

In the reaction

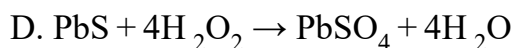
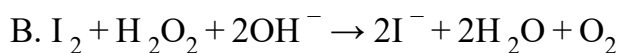
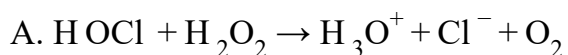


Question 79

In which of the following reactions, hydrogen peroxide acts as an oxidizing agent?

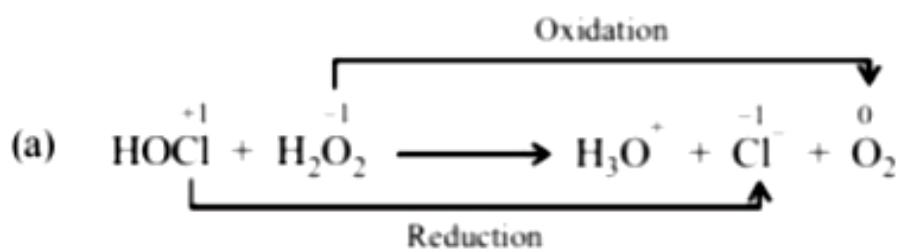
[Online April 8, 2017]

Options:



Answer: D

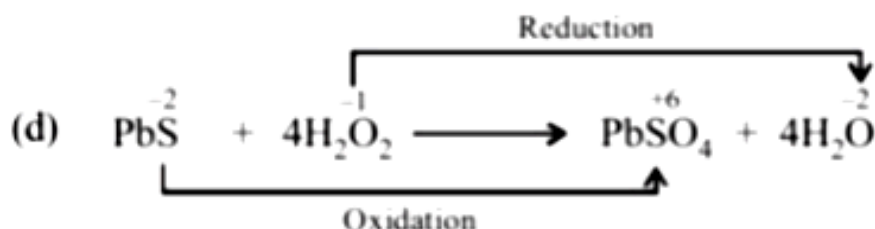
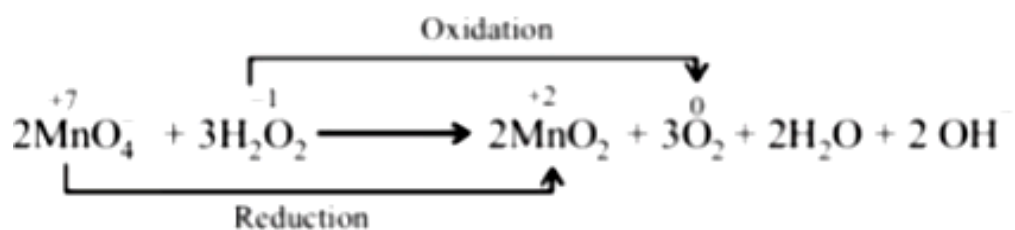
Solution:



(b)



(c)



Notice that the oxidation state of oxygen goes from -1 on the H_2O_2 to -2 on the H_2O means H_2O_2 is being reduced. On the other hand the oxidation state of sulfur is going from -2 on the PbS to $+6$ on the PbSO_4 . i.e Sulfur is being oxidised

Question80

Copper becomes green when exposed to moist air for a long period. This is due to:

[Online April 12,2014]

Options:

- A. the formation of a layer of cupric oxide on the surface of copper.
- B. the formation of a layer of basic carbonate of copper on the surface of copper.
- C. the formation of a layer of cupric hydroxide on the surface of copper.
- D. the formation of basic copper sulphate layer on the surface of the metal.

Answer: B

Solution:

Copper when exposed to moist air having CO_2 . It gets superficially coated with a green layer of basic carbonate $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$

Question81

**Amongst the following, identify the species with an atom in +6 oxidation state:
[Online April 19,2014]**

Options:

- A. $[\text{MnO}_4]^-$
- B. $[\text{Cr(CN)}_6]^{3-}$
- C. Cr_2O_3
- D. CrO_2Cl_2

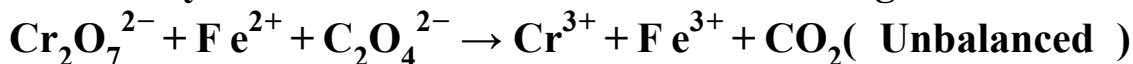
Answer: D

Solution:

CrO_2Cl_2
Let O. No. of Cr = x
 $\therefore x + 2(-2) + 2(-1) = 0$
 $x - 4 - 2 = 0$
 $\therefore x = +6$

Question82

How many electrons are involved in the following redox reaction?



[Online April 19, 2014]

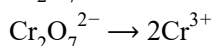
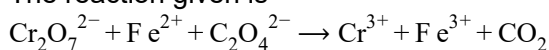
Options:

- A. 3
- B. 4
- C. 6
- D. 5

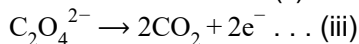
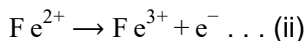
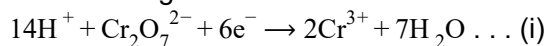
Answer: A

Solution:

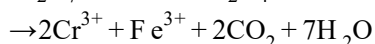
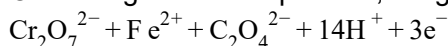
The reaction given is



On balancing



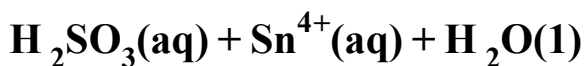
On adding all three equations, we get



Hence the total no. of electrons involved in the reaction = 3

Question83

Consider the reaction:



[Online April 19, 2014]

Options:

- A. Sn^{4+} is the oxidizing agent because it undergoes oxidation
- B. Sn^{4+} is the reducing agent because it undergoes oxidation

C. H_2SO_3 is the reducing agent because it undergoes oxidation

D. H_2SO_3 is the reducing agent because it undergoes reduction

Answer: C

Solution:

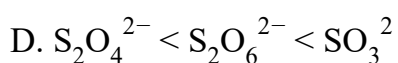
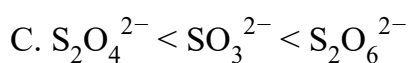
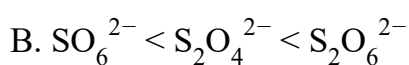
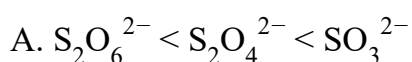
If an electronegative element is in its lowest possible oxidation state in a compound or in free state. It can function as a powerful reducing agent. e.g. I^-

Question84

Oxidation state of sulphur in anions SO_3^{2-} , $\text{S}_2\text{O}_4^{2-}$ and $\text{S}_2\text{O}_6^{2-}$ increases in the orders :

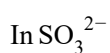
[Online April 22, 2013]

Options:

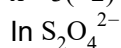


Answer: C

Solution:



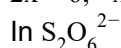
$$x + 3(-2) = -2; x = +4$$



$$2x + 4(-2) = -2$$

$$2x - 8 = -2$$

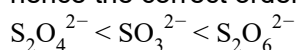
$$2x = 6; x = +3$$



$$2x + 6(-2) = -2$$

$$2x = 10; x = +5$$

hence the correct order is



Question85

Given:



**The values of X, Y and Z in the above redox reaction are respectively :
[Online April 9, 2013]**

Options:

A. 2, 1, 2

B. 2, 1, 3

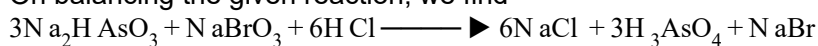
C. 3, 1, 6

D. 3, 1, 4

Answer: C

Solution:

On balancing the given reaction, we find



Question86

**Which one of the following cannot function as an oxidising agent?
[Online April 25, 2013]**

Options:

A. I^-

B. S(s)

C. $\text{N O}_3^-(\text{aq})$

D. $\text{Cr}_2\text{O}_7^{2-}$

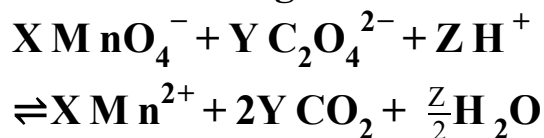
Answer: A

Solution:

If an electronegative element is in its lowest possible oxidation state in a compound or in free state. It can function as a powerful reducing agent. e.g. I^-

Question87

In the following balanced reaction,



values of X, Y and Z respectively are
[Online May 12, 2012; 2013]

Options:

A. 2, 5, 16

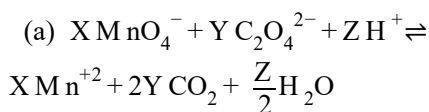
B. 8, 2, 5

C. 5, 2, 16

D. 5, 8, 4

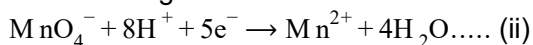
Answer: A

Solution:



First half reaction $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$ (i)

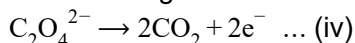
On balancing



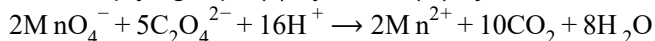
Second half reaction



On balancing



On multiplying eqn. (ii) by 5 and (iv) by 2 and then adding we get

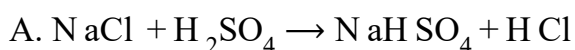


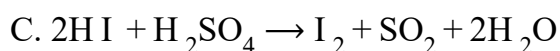
Question88

Which of the following chemical reactions depict the oxidizing behavior of H_2SO_4 ?

[2006]

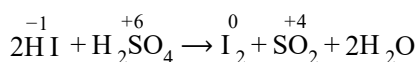
Options:





Answer: C

Solution:



in this reaction oxidation number of S is decreasing from + 6 to +4 hence undergoing reduction and for HI oxidation number of I is increasing from -1 to 0 hence undergoing oxidation, therefore H_2SO_4 is acting as oxidising agent.

Question89

Several blocks of magnesium are fixed to the bottom of a ship to [2003]

Options:

- A. make the ship lighter
- B. prevent action of water and salt
- C. prevent puncturing by under-sea rocks
- D. keep away the sharks

Answer: B

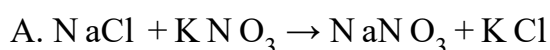
Solution:

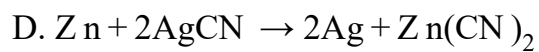
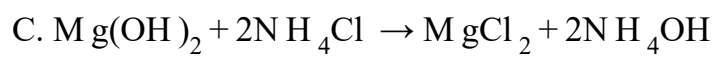
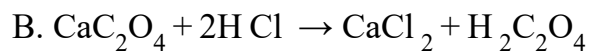
Magnesium provides cathodic protection and prevents rusting or corrosion.

Question90

Which of the following is a redox reaction? [2002]

Options:





Answer: D

Solution:

